

- 6 Fig. 6.1 shows a resistor network connected to a 12 V battery of negligible internal resistance. A voltmeter of resistance $6.0\text{ k}\Omega$ is connected in parallel across the $200\text{ }\Omega$ resistor. When the light intensity is varied, the resistance of the light-dependent resistor (LDR) ranges from $120\text{ }\Omega$ to $500\text{ }\Omega$.

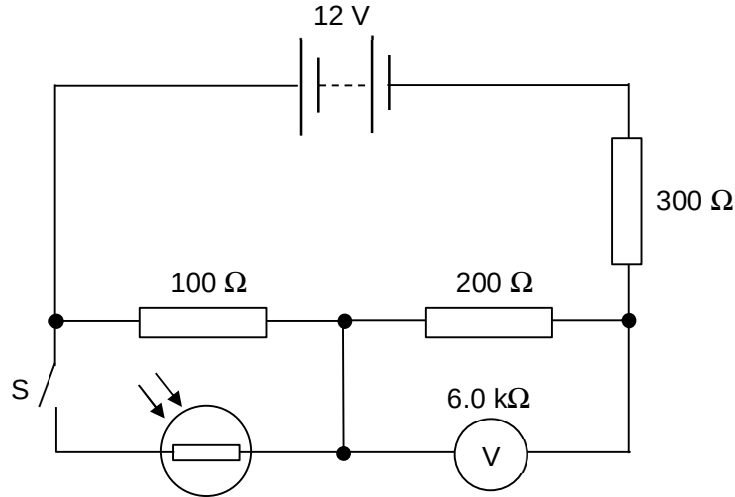


Fig. 6.1

(a) Switch S is open.

(i) Calculate the total resistance of the circuit.

total resistance = Ω [2]

(ii) Calculate the current from the battery.

current = A [1]

(iii) Hence or otherwise show that the reading on the voltmeter is 3.9 V.

[2]

(b) Switch S is now closed. The light intensity is at a maximum.

Calculate the power supplied to the $200\ \Omega$ resistor.

power = W [3]

